

Canopy water content is related to variability in leaf water over time but not across space in an oak savanna ecosystem

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ABSTRACT

Declines in canopy water content (CWC) derived from visible/near-infrared imaging spectroscopy data have been linked to tree water stress and mortality, suggesting that CWC could be a valuable tool for monitoring ecological drought. CWC is sensitive to both the area of leaves in a tree canopy (leaf area index, LAI) and to the relative proportions of water mass and dry mass in the leaf (leaf water content, LWC). However, the relative contribution of each of these properties to CWC is not well understood. As a result, any single CWC measurement is likely underdetermined and challenging to reliably link to specific tree drought responses. Here we leverage a first-of-its-kind high-frequency imaging spectroscopy time series to explore how comparatively sensitive CWC is to LAI and LWC across both time and space in an oak savanna. Coincident with four imaging spectroscopy data acquisition flights, we measured LAI and LWC, as well as leaf water potential (a diagnostic of tree hydraulic stress). We found that LAI was the dominant control on CWC ($R^2 = 0.11\text{--}0.25$), and that across space, CWC shows no sensitivity to LWC ($R^2 = 0.01\text{--}0.18$). However, as tree water stress increased over time and LWC declined, CWC declined in step (average LWC vs. average CWC over time yield $R^2 = 0.89$). Furthermore, we found that more negative leaf water potentials were associated with low LWC and reduced LAI, both across time and space, and therefore low CWC. Altogether, these results suggest that the main utility of CWC for monitoring tree water stress comes from its status as an integrated measure of both LAI and LWC, each of which may show coordinated or independent responses to drought that can differ across space and time.

1. Introduction

Increases in the frequency of severe droughts brought about by climate change have instigated elevated tree mortality in ecosystems globally over the past several decades (Bigler et al., 2006; van Mantgem et al., 2009; Hammond et al., 2022), with negative ramifications for biodiversity (Clark et al., 2010; Clark et al., 2016), fire risk (Hanson and Weltzin, 2000; Bigler and Veblen, 2011), and the terrestrial carbon sink (Kurz et al., 2008; Adams et al., 2012; Anderegg et al., 2020; Liu et al., 2023). However, despite advances in our understanding of the mechanisms underlying tree drought mortality, our ability to monitor and forecast the effects of drought on natural ecosystems remains limited (Trugman et al., 2021; Venturas et al., 2021). In part, this is due to a gap in data availability: Field surveys of ecosystem water status are necessarily limited in scope due to their time-intensive nature, and any in-situ

study of the physiological mechanisms of tree water stress is contingent on early identification of vulnerable trees, which is often not possible.

Visible/near-infrared imaging spectroscopy data, collected via aircraft or satellite, have the potential to revolutionize monitoring of ecosystem responses to drought stress by collecting measurements of plant traits and biochemistry over large, continuous spatial and temporal scales that would be infeasible to attempt for field measurements (Cawse-Nicholson et al., 2021). One emerging diagnostic for monitoring tree mortality risk from remote sensing is canopy water content (CWC), a measurement of the amount of liquid water contained in vegetation per unit ground area that can be derived from imaging spectroscopy data. CWC is a well-established tool within remote sensing for measuring live fuel moisture (Dennison et al., 2003; Maki et al., 2004; Sow et al., 2013) and vegetation biomass (Roberts et al., 1997; Ustin et al., 1998; Roberts et al., 2004; Cheng et al., 2008), but recent

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expansion in availability of imaging spectroscopy data has increased interest in connecting CWC to tree water stress and subsequent mortality risk (Baeza et al., 2021; Le et al., 2023). Several studies have demonstrated a connection between decreases in CWC and elevated branch dieback or tree mortality rates (Asner et al., 2015; Brodrick and Asner, 2017; Martin et al., 2018; Brodrick et al., 2019), suggesting that changes in CWC through time could be a valuable tool for ecosystem monitoring. However, the connection between CWC and tree water stress cannot be fully exploited without a more thorough understanding of both the physiological and the optical mechanisms that drive this relationship.

Diagnosing changes in CWC as specific physiological drought responses presents a challenge because diverse tree drought responses interact with CWC in nonlinear ways. Specifically, CWC is generally hypothesized to be an integrated signal of total canopy leaf biomass or leaf area index (LAI), leaf mass per area, and the amount of water in the leaves, which is often expressed as a ratio between the water mass and dry mass of the leaf as leaf water content (LWC). CWC has been found to be a useful proxy measurement for LAI (Asner et al., 2004; Roberts et al., 2004), which can vary strongly across space and time in response to differences in environmental conditions and species composition. Importantly, one drought avoidance strategy that trees experiencing water stress can employ is reducing LAI, either by growing fewer or smaller leaves (Löf and Welandar, 2000; Pedrol et al., 2000; Otieno et al., 2005; Meier and Leuschner, 2008) or by shedding leaves prematurely, even in trees not traditionally considered to be drought deciduous (Munné-Bosch and Alegre, 2004; Carnicer et al., 2011). LWC can change diurnally with stomatal responses to diurnal cycles in radiation and climate, seasonally as a function of leaf development and dry matter accumulation (which affects leaf mass per area) (Boving et al., 2025), and/or sub seasonally in response to variability in soil moisture or atmospheric evaporative demand (Hall and Kaufmann, 1975; Tyree and Sperry, 1988; Matthews et al., 2017; Martinez-Vilalta et al., 2018; Nolan et al., 2020). Because of the potential for both leaf- and canopy-level responses, translating changes in CWC to ecologically relevant indicators of tree drought response (i.e., a change in LAI and/or LWC) is an under constrained problem. Furthermore, relationships between CWC and tree water status that hold across space may not translate well to time series data due to the complexity added by phenological cycles, changes in leaf morphology, allocation, and differential drought responses across environmental gradients (Boving et al., 2025).

Further complicating the relationship between CWC and tree stress is the fact that LWC is not on its own mechanistically related to tree water stress. Low LWC is indicative of elevated water stress primarily through its relationship with leaf water potential (LWP), a measure of the tensile forces that leaf water exerts on the plant cell walls. As LWC declines, LWP generally grows increasingly negative as these tensile forces increase in strength, which—in extreme cases—can induce embolism in leaf and stem xylem by pulling in air from outside the xylem elements. Embolized xylem elements within the plant are often considered to be irreversibly damaged (Hacke et al., 2001). As a result, embolism is one of the main mechanisms by which water stress damages trees. Therefore, LWP (which relates directly to the risk of embolism within the plant) is a mechanistic diagnostic of tree water stress (Sperry et al., 2002; Ander-egg et al., 2016). However, the relationship between LWC and LWP is complex: The two measurements are related through a piecewise, nonlinear relationship (Kennedy and Booth, 1958) that varies strongly across species and can even vary significantly across individuals within a species due to differences in leaf anatomy, water stress history, and solutes contained within the leaf water (Sancho-Knapik et al., 2010; Bartlett et al., 2012; Boving et al., 2025). Given that CWC measures water as a mass per unit area rather than as a potential, it is likely to be more strongly correlated with LWC and other mass- or area-based leaf water metrics than with LWP, and CWC's relationship with actual tree water stress should be similarly complex. However, few studies exist that explore how CWC relates to LWP (Boving et al., 2025; Martin et al., 2018).

In 2022 an airborne imaging campaign, the SBG High-Frequency Timeseries (SHIFT), (Chadwick et al., 2025) flew a series of imaging spectroscopy data collection flights over Santa Barbara County, California on a near-weekly basis during a spring green up from February through May, and two end-of-season imaging flights in September at 5 m spatial resolution. The SHIFT data acquisition area included a large oak-grass savanna dominated by multiple species of native oak trees that have exhibited elevated water stress and mortality in recent years associated with a prolonged decadal drought (Andreadis et al., 2005; Williams et al., 2020). The airborne measurements captured the progression from wet season to dry season, providing a natural experiment to compare CWC and LWC in step as water stress increased during the dry Mediterranean summer in an already water-stressed ecosystem. To our knowledge, the SHIFT imaging campaign was the first set of measurements of its kind to record changes in CWC with high-frequency, sub-seasonal variations in water availability over a single year. Thus, SHIFT provides an unprecedented opportunity to quantify what drives variations in CWC across space and time in an ecosystem where individual tree crowns can be easily delineated and changes in organismal-level stress may be detected. Concurrent with four of the SHIFT flights, we collected ground-based measurements of LWP, LWC, and LAI for two dominant tree species (*Quercus agrifolia* and *Quercus douglasii*) across a topographic gradient to compare with CWC. This dataset enables us to measure how LWC and LAI contribute to a CWC measurement, and if this relationship varies over time. In addition, by comparing CWC with LWP, we can assess how effectively CWC may be used to map tree water stress, and identify the mechanisms that underly those relationships. We asked: (1) How do CWC measurements derived from imaging spectroscopy relate to in situ measurements of LAI and LWC, and how do the wavelengths used to calculate CWC influence this relationship? (2) Do relationships between CWC and LWC or tree stress (measured through LWP) differ across space versus over time? (3) What vegetation properties dictate the relationship between CWC and tree water stress?

2. Methods

2.1. Study area

Sedgwick Reserve is a semi-arid oak savanna located in southern California (Fig. 1a-b). The climate is Mediterranean, with a mild winter wet season and a hot summer dry season. The landscape is dominated by three endemic oak species: blue oak (*Quercus douglasii*), coast live oak (*Quercus agrifolia*), and valley oak (*Quercus lobata*), with an understory of annual grasses. In this study, we collected data for *Q. douglasii* ($n = 42$) and *Q. agrifolia* ($n = 14$), the more abundant of the three oak species. Each species exhibits distinct phenological strategies: *Q. douglasii* is a temperate deciduous species, while *Q. agrifolia* is evergreen, with new leaves growing in the spring and lasting up to two years. Both species often co-occur and exist across a variety of environmental conditions, including differences in elevation, slope, aspect, and soil composition. In this study, we created a representative dataset of water status for trees at Sedgwick Reserve by designing our data collection protocol around these natural gradients in water availability, collecting measurements for individual trees across four distinct locations (Fig. 1b). Soils at the site are clayey-skeletal, mixed, active, thermic Ultic Palexerolls, with northernmost sampling site located on fine-loamy, mixed, superactive, thermic Pachic Argixerolls (Soil Survey Staff, 2022). These four sites sample trees growing in a variety of soil textures with a range of clay (22.5–47.5%) and sand (6.7–57%) fractions, as well as a variety of topographic positions, including hillslopes, ridges, and valleys. Together, these factors provided a gradient of water stress conditions for both tree species.

The study year, 2022, was a drier than average year at Sedgwick Reserve, with a cumulative rainfall of 231 mm (in contrast with the annual average of 322 mm) (Fig. 1c). In a normal year for this

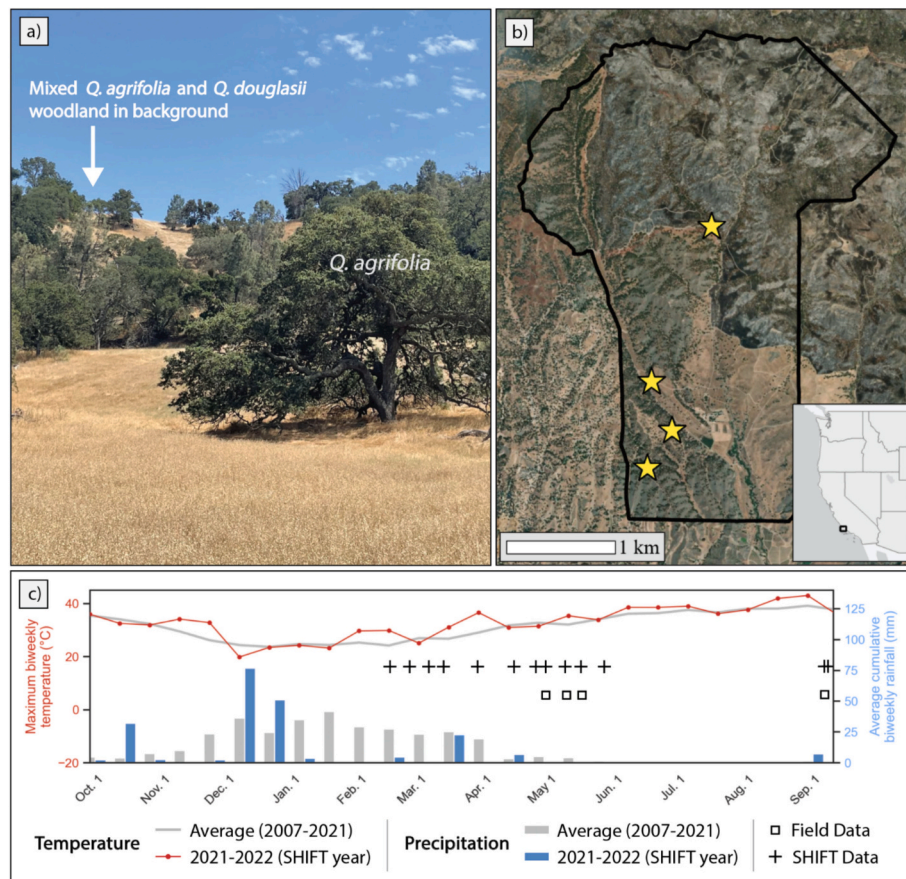


Fig. 1. The study site, located at Sedgwick Reserve, is a heterogeneous patchwork of *Q. douglasii* and *Q. agrifolia* woodlands and large, isolated *Q. lobata* and *Q. agrifolia*, both with an understory of annual grasses and shrubs. (a) Ground-level photograph of the study site. (b) Overhead view of Sedgwick Reserve (reserve boundary outlined in black), with data collection sites marked as stars. Base map imagery provided by Esri (Esri, 2024). (c) Climatograph showing the average maximum biweekly temperature and the cumulative biweekly precipitation for a recent historical average (2007–2021 water years) and for the study year (2021–2022 water year). Meteorological data provided by Innovative Datasets for Environmental Analysis by Students at <https://ideas.geog.ucsb.edu/> (Roberts et al., 2010).

ecosystem, tree leaf growth occurs after moderate rain in late winter and spring, and plant water stress increases naturally as spring progresses to summer, soil water content decreases, and atmospheric vapor pressure deficit increases. In contrast, in 2022, most of the light rains fell in the preceding December (2021) and the rest of the year was almost completely dry. A dry water year, combined with the legacy effects of nearly two decades of drought in California (Williams et al., 2020), would be expected to result in moderate to high water stress for nearly all trees in the study domain – even early in the dry season – regardless of topographic position or species. This means that we expected LWP, LWC, LAI, and (most likely) CWC to be lower than average.

2.2. Imagery processing and canopy water content calculation

The SHIFT measurement campaign (Chadwick et al., 2025) included 13 overflights by the Airborne Visible InfraRed Imaging Spectrometer - Next Generation (AVIRIS-NG) instrument (Hamlin et al., 2011), including 11 springtime flights and 2 autumn flights (Fig. 1c). AVIRIS-NG as an instrument is well-suited for collecting accurate and intercomparable CWC measurements: Its high spectral resolution (5 nm $\pm$ 0.5 nm) provides many surface reflectance measurements in wavelength ranges sensitive to liquid water, and its relatively narrow field of view ($36 \pm 2^\circ$) minimizes cross-track signal variability caused by differences in viewing geometry. Data were corrected from radiance to surface reflectance using the Imaging Spectrometer Optimal FITting (ISOFIT) package v.2.9.9.2 (Thompson et al., 2018; Brodrick et al.,

2025), an open-source software package developed for fitting imaging spectrometer data to surface and atmospheric optical models using a Bayesian framework.

All imagery was collected with the same flight pattern at a target spatial resolution of 5 m with subpixel geolocation accuracy (Chadwick et al., 2025), ensuring comparability between measurements across the time series. This pixel size roughly corresponds to a medium-sized tree crown within the scene. Trees with subpixel crown sizes, particularly *Q. douglasii*, are normally tightly clustered together into a relatively closed canopy (Fig. 1a) and therefore still provide a good signal on a per-pixel basis. For each tree in our study dataset, we extracted the spectral measurement that was closest in space to the base of the trunk of the tree using GPS measurements collected with a Trimble R8 Model 2 GPS device. For trees located in the overlap between two flightlines, we preferentially used data from the “back scattering” viewing geometry where the instrument is positioned on the same side of the target as the sun. This should reduce the impact of shadows, resulting in an overall higher signal returned to the instrument (Liu et al., 2024) and, in turn, a more sensitive CWC measurement. However, we found that viewing geometry had a relatively small impact on estimated CWC values; thus, this decision likely did not affect the final results (Fig. S1).

CWC was calculated using a modified Beer-Lambert inversion model implemented within ISOFIT package (invert_liquid_water function). This approach, which is commonly used for calculating CWC from surface reflectance data (e.g., Miller et al., 2022), offers the advantage of being less computationally intensive than other methods (such as inverting

radiative transfer models) and is therefore more suitable for analyses that cover large geographic extents. According to the Beer-Lambert Law, the amount of light transmitted through a material varies with the inverse exponential of the wavelength-specific absorption coefficient of that material (α , a measured constant that varies with wavelength λ) and the distance traveled through that material (in this case, the amount of liquid water in the tree canopy or CWC). This transmitted radiation can ultimately be scattered or reflected off the surface to be measured by a sensor as a surface reflectance ρ_λ :

$$\rho_\lambda = e^{-\alpha_\lambda \cdot \text{CWC}} \quad (1)$$

The Beer-Lambert CWC retrieval implemented in ISOFIT and used in this study deviates only slightly from approaches that have been previously described in that it includes an affine adjustment, which accounts for the effects of light absorbed by materials other than water, such as branches, soil, senesced vegetation, or shadows:

$$\rho_\lambda = (a + b\lambda) e^{-\alpha_\lambda \cdot \text{CWC}} \quad (2)$$

where a and b are empirical parameters used to describe the affine fit. Using values for α_λ from Kedenburg et al., 2012, the invert_liquid_water submodule estimates values for a , b , and CWC that maximize the fit between a measured and modeled surface reflectance ρ_λ .

The wavelengths used to calculate CWC can substantially influence the accuracy of the retrieved values, yet most studies use wavelengths from 850 to 1100 nm (Green et al., 2006; Bohn et al., 2020). Scattering of light within tree canopies introduces deviations from the Beer-Lambert model that become non-negligible under certain conditions. Because the magnitude of these scattering effects varies with wavelength, the CWC values retrieved using the Beer-Lambert approach also vary accordingly, potentially affecting the relationship between remotely retrieved CWC data and field measurements. To quantify and account for this effect, we calculated CWC by inverting all possible wavelength ranges between 850 and 1350 nm that included two or more bands and compared the magnitude of retrieved CWC values across wavelength ranges. Next, to determine which wavelength range was most appropriate for our study area, we regressed these remote CWC measurements against a field validation of CWC, measured as (LAI) $\times$ (Leaf water per area). Based on the goodness of fit between the CWC data and the field validation, we selected an appropriate wavelength range to use for our study site.

2.3. Field measurements

For a summary of field measurements and the initialisms used to refer to them through the text, see Table 1.

2.3.1. LAI

LAI was measured using a METER LP-80 ceptometer, which works by comparing above canopy radiation (measured using a nearby Meter Apogee PAR sensor) to below canopy radiation (measured in situ with the ceptometer). Approximately 15–20 ceptometer measurements were collected in full sun conditions within 2 h of solar noon in the understory of each tree in varying positions and orientations, which were then averaged together to provide an overall estimate of below canopy radiation. LAI was then calculated from these measurements using materials provided by METER, assuming a spherical leaf angle distribution

Table 1
Definition and units of key abbreviations used throughout.

Abbreviation	Full Name	Units
CWC	Canopy Water Content, measured by AVIRIS-NG	g/cm ²
LWC	Leaf Water Content, measured in situ	g/g
LAI	Leaf Area Index, measured in situ	m ² /m ²
LWP	Leaf Water Potential, measured in situ	MPa

(Campbell, 2023). While it's unlikely that the leaf angle distribution in these trees is exactly spherical, the sensitivity of LAI measurements to this assumption was found to be low: Calculating LAI using most and least spherical distributions observed previously in other oak trees (Kull et al., 1999) yielded an average difference in estimated LAI of 0.03 m²/m², a small error relative to the range of LAI values observed here. To fully capture the wide variability in LAI across the landscape, we also collected LAI for 42 additional *Q. douglasii* (total $n = 84$) and 3 additional *Q. agrifolia* trees (total $n = 17$). We collected four census measurements of LAI for all trees in this larger dataset, spread out across the measurement dates. Some LAI censuses took multiple days to complete. Thus, LAI measurements were aggregated into four groups: “Late April,” “Early May,” “Late May,” and “September” (Fig. S3).

To determine if and how much LAI changed over the study period, we conducted a repeated measures analysis of variance (ANOVA) using the statsmodels python package (Seabold and Perktold, 2010) as well as pairwise tests (Table S2) using the python package pingouin (Vallat, 2018) to quantify statistical significance. We determined that LAI was relatively stable for all measurement dates except for the final date in September (ANOVA yielded F-statistic = 5.6514; p -value = 0.001; pairwise tests yielded one statistically significant difference between “Early May” and “September” measurements). Therefore, in order to leverage this period of phenological stability and minimize measurement noise, we averaged together the “Late April,” “Early May,” and “Late May” LAI measurements for comparison with the springtime CWC measurements and isolated the “September” measurements for comparison with the September CWC dataset. These LAI data were compared with the CWC measurements using linear regressions. In general, CWC is understood to vary approximately with the square root of LAI (e.g., Roberts et al., 2004); however, for the relatively low LAI values observed here (all <4.5 m²/m²), we expected a linear relationship (e.g., Roberts et al., 1998). As the LAI dataset was larger than the LWC dataset, the LAI data was randomly subsampled 100 times to datasets equal in size to the LWC dataset. Statistical metrics (R^2 , p -value) were then calculated for each subsample and averaged together for a final result.

It should be noted that the LAI estimated using this method is likely a moderate overestimate of the actual value because the measurement includes light attenuation from both leaves and woody material. In this study, we assume that this LAI estimate is well-correlated with total leaf area that would be obtained if it were possible to only measure the light attenuation caused by leaves. This assumption is partially supported by previous research showing that large branches are often shaded by leaves, and therefore intercept less light and contribute less to a below-canopy attenuation than might be expected (Kucharik et al., 1998). However, the oak species studied here have a large proportion of woody biomass and the within-canopy light scattering environment may differ from what has been tested previously. To mitigate this issue, we compare LAI and CWC using only linear regressions with adjustable intercepts, which should account for this probable high bias in our LAI measurements.

2.3.2. LWC

To constrain how CWC varied with LWC across both space and time, we collected LWC data within three days of the AVIRIS-NG overflight (Fig. 1c). Leaf collection for LWC measurement was timed similarly to the image acquisition, namely within two hours of solar noon. LWC (sometimes also referred to as live fuel moisture in fire research) is measured as a ratio of the mass of water in a leaf to the dry mass of the leaf material itself:

$$\text{LWC} = (\text{Wet mass} - \text{Dry mass}) / (\text{Dry mass}) \quad (3)$$

Approximately 6–9 sun leaves for each tree in the study set were collected at midday and weighed to record the wet mass. Leaves were then placed in a drying oven for 48 h at 60 °C and weighed again to calculate the dry mass. To assess how CWC relates to LWC, these

measurements were compared to the CWC data across space through a series of linear regressions, comparing the LWC data to the CWC measurements for those points in space for each corresponding date. Similarly, we calculated the relationship between LWC and CWC across time using a linear regression between the average CWC and the average LWC for all trees in the study data set across the four measurement dates.

To minimize the effects of phenology on our results, springtime LWC measurements were collected in late April through late May. From February through mid-April, CWC was likely an integrated measurement of water stored in both trees and understory grasses, which could confound our ability to relate CWC to measurements of tree structure and hydration. However, by late April, all understory grasses at our data collection sites had senesced (Fig. S2, S3), and CWC measurements from pixels containing only grasses were uniformly zero g/cm². As a result, over our LWC data collection period, the vast majority of the CWC signal should be associated with water contained in oak tree leaves and not understory grasses or shrubs. Furthermore, this data collection window targets a period where leaf growth was complete or nearly complete for many of our study trees: Leaf mass per area can change substantially during leaf development, which has been found to confound the relationship between CWC and LWC in this ecosystem (Boving et al., 2025). By focusing on this late-spring period of relative phenological stability, we limit the effects of this and other potential confounding variables.

For the purposes of validating our CWC measurements, we also measured average leaf area for three dates (4/29/2022, 5/11/2022, and 9/14/2022) using leaves collected for measuring LWC. Leaf areas were measured in ImageJ (Schneider et al., 2012) and combined with water masses calculated above (wet mass – dry mass) to provide a measurement of the average mass of leaf water per unit area. This was then multiplied by LAI to provide an in situ estimate of the “true” CWC for that tree that could be compared with airborne measurements.

2.3.3. Midday and Predawn LWP

Finally, to understand how CWC measurements relate to physiologically relevant metrics of water stress in the tree, we collected measurements of LWP at both predawn and midday for the same four dates. There are two distinct LWP measurements that are useful for diagnosing physiological stress: midday LWP, when trees are expected to be experiencing peak water stress conditions, and predawn LWP, when water in leaves is approximately equilibrated with soil water (Ritchie and Hinckley, 1975). In this way, predawn LWP can be used to measure a tree's baseline stress and soil water access, while midday LWP provides an integrated measurement of atmospheric demand, stomatal conductance, hydraulic efficiency, and soil water stresses and is indicative of peak stress conditions encountered by the tree during that day (Martinez-Vilalta et al., 2018).

LWP was measured for trees at the study site using a Scholander pressure bomb (Scholander et al., 1965) for sun leaves at midday (within 1.5 h of solar noon) and at predawn (1–3 h before sunrise). Samples were collected from the mid- to upper canopy by cutting a large branch and then immediately cutting 5–9 leaves at the base of the petiole, quickly wrapping the leaves in tinfoil (for midday measurements only), bagging the leaves in a whirl pack back bag, and then placing the whirl pack in a ziplock bag containing a moist paper towel. Leaves were allowed to equilibrate and then measured between 30 min and 6 h of collection (Rodriguez-Dominguez et al., 2022). Similar to LWC, midday LWP measurements were collected in early spring and in September within three days of data collection by AVIRIS-NG, while predawn LWP measurements were collected within eight days, given that soil drying typically occurs on the scale of weeks. This provided a total of four dates with sufficient LWC, LWP, and CWC data to make a robust comparison. We performed the same comparisons between the LWP data and the CWC data as we did the LWC data; specifically, both midday and predawn LWP were regressed against CWC across space for each SHIFT date (capturing tree-to-tree variability across space) with species as a covariate, as well as in aggregate as averages for each measurement date

(capturing the relationship across time).

2.4. Measuring controls on tree water stress on the landscape

To determine the dominant controls on tree water stress across the study area, we ran a series of linear mixed effects models predicting field measurements from topographic variables. Topographic variables, including elevation, slope, and aspect were calculated using an airborne lidar dataset collected at Sedgwick Reserve in 2020 (Brandt, 2021) and were included as fixed effects, with measurement date and species as a random intercept term. To increase suitability for linear modeling, aspect was reformulated to a northness term calculated as the cosine of aspect. All predictors were centered and scaled in preparation for linear modeling. All three water status metrics used in this study (LWC, predawn LWP, and midday LWP) were considered separately as response variables.

Mixed effects models were built using the Python package statsmodels (Seabold and Perktold, 2010). For each possible combination of topographic effects and response variables, we fit three mixed models: (a) a topography only model, (b) a topography and CWC model, and (c) an interaction model that included additive contributions from CWC as well as one-way interactions with all topographic variables. Next, for each response variable, we selected the most parsimonious set of predictor variables using the Akaike Information Criterion (AIC) (Akaike, 1973) for model selection. We fit all subset models of the full model and selected the model with the lowest AIC. If a simpler model than the model that minimized AIC was <2 from the model that minimized AIC (used as a general AIC cutoff for what constitutes a significant model improvement, Zuur et al., 2010), we selected the simpler model instead.

Finally, to determine how informative CWC was for mapping tree water stress on a landscape level, we compared the best topography-only model selected for each response variable to the corresponding models that included CWC and analyzed the associated change in AIC and change in marginal R² value.

3. Results

3.1. CWC values vary by wavelength region

CWC measurements calculated from the AVIRIS-NG imagery varied substantially depending on the wavelengths used for the inversion (Fig. 2a). Measurements that used only shorter wavelengths, such as the 850–1150 nm wavelength range, generally resulted in lower CWC values when compared with measurements that also used longer wavelengths, such as the 1100–1300 nm wavelength range. Notably, CWC values calculated from the remotely sensed imagery were nearly an order of magnitude higher than the in situ estimated leaf water per unit area, which ranged from 0.012 to 0.063 g/cm².

The strength of the relationship between CWC and the estimated leaf water per unit area measured on the ground also varied depending on the wavelength region used (Fig. 2b). In general, the strongest correlations were observed using wavelengths that fully contain the water absorption feature centered at 1210 nm (approximately 1100–1300 nm). Lower correlations were observed at wavelengths that contain both liquid water absorption features (approximately 850–1300 nm) or only the first liquid water absorption feature centered at 970 nm (approximately 850–1100 nm). This difference was primarily driven by failed convergence in the CWC inversion, resulting in a measurement of 0 g/cm² for locations with a nonzero CWC value measured in situ, which was more common when CWC was calculated using wavelengths where the absorption coefficient of water is lower. Considering these results, for the rest of the paper we focus on using CWC values retrieved using a 1100–1300 nm wavelength range.

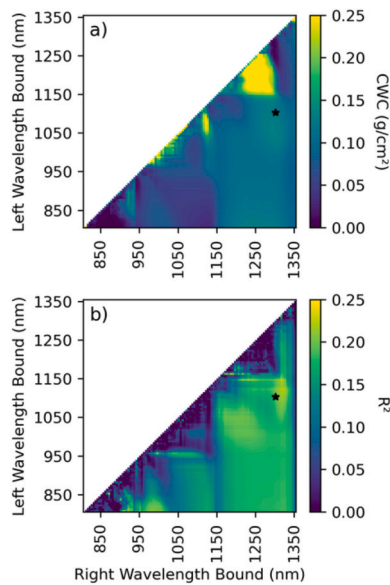


Fig. 2. The CWC values retrieved vary depending on the wavelength range used for the inversion, which induces differences in goodness of fit between the CWC as measured remotely and as measured in situ with field measurements. (a) Average CWC value across all trees for an arbitrarily chosen date (4/29/2022). (b) R^2 values of linear regressions between the CWC values calculated at those wavelengths and the in situ CWC validation data based on field measurements. Stars indicate wavelengths used in all further analyses in this paper.

3.2. CWC is more sensitive to leaf area index than leaf water content

For all four measurement dates, CWC as measured from AVIRIS-NG and field measurements of LAI were more strongly correlated than CWC and field measurements of LWC (Fig. 3). Across the full dataset, the relationship between LAI and CWC was consistently positive (p -value < 0.01 with a positive slope on all dates). In contrast, the CWC-LWC relationship exhibited slopes of varying magnitudes ranging from positive to negative, and the only statistically significant relationship showed an increase in CWC as LWC as measured in the field declined – the opposite of physical expectations (Fig. 3b). LAI datasets subsampled to be of equal size to the LWC data yielded two dates with statistically significant results. Altogether, these results provide good evidence that, within any individual remotely sensed scene, CWC is primarily sensitive to LAI and not LWC.

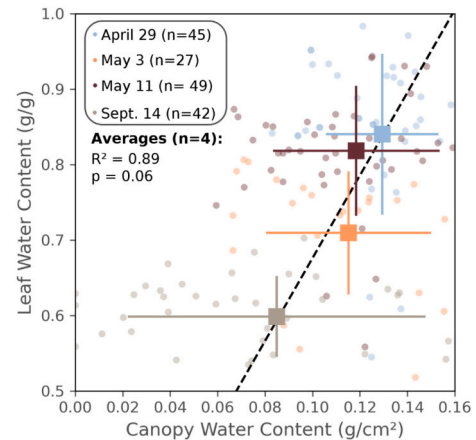


Fig. 4. Across the four measurement dates, average CWC across all trees was correlated with average LWC. Scatterplot showing how the average LWC across all trees for each date in the study dataset relates to the average CWC for that date. Circular points indicate individual tree measurements, while the large squares show the average for that measurement date. Error bars are sized to one standard deviation for the respective measurements on that date.

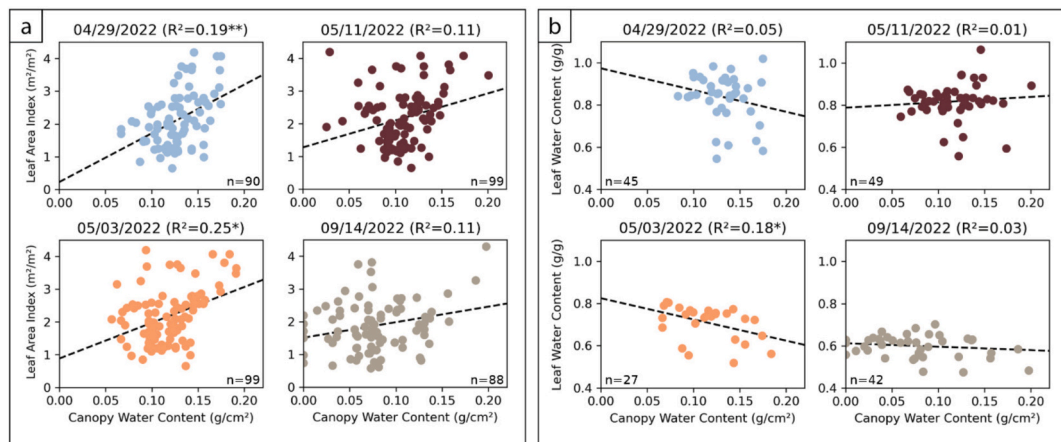


Fig. 3. The relationship between CWC and LAI is weak ($R^2 < 0.2$), but consistently positive across measurement dates. In contrast, the relationship between CWC and LWC is variable in direction. (a) Linear regressions between LAI and CWC for four of the SHIFT measurement dates. R^2 values and p -values calculated using the LAI data indicate subsampled models that are of the same sample size as the LWC data. (b) Same as (a), but for LWC instead of LAI. For both (a-b), * indicates results that were statistically significant at $p < 0.05$; ** indicates results that were statistically significant at $p < 0.01$.

time (e.g., more overpasses of the same location) or across space (e.g., averaging several pixels across space).

3.3. Relating CWC to tree water stress

The study year was much drier than average, with rainfall 28% lower than the water year average; as a result, nearly all trees in the study data set exhibited moderate to high levels of water stress through the entire study period. Average midday LWP was below -2 MPa for all four measurement dates, and by September some trees reached midday LWP of -5 , values beyond the water potential that causes 50% xylem embolism for both species (Skelton et al., 2018) indicating fairly extreme stress conditions. This likely resulted in lower LWP, LWC, CWC, and potentially LAI values than could be expected in a wetter year.

Across space, the relationship between midday LWP and CWC was more consistent (and in the right direction) than the LWC and CWC relationship. Midday LWP and LWC have a strong but nonlinear and variable relationship (Boving et al., 2025), so overall similarity of results is expected. Within any single scene, CWC generally exhibited a weak or null correlation with midday LWP ($R^2 \leq 0.3$) (Fig. 5a), with two dates where the relationship was statistically significant and positive. In contrast, across time, the average CWC was correlated with the average

midday LWP (Fig. 5b; $R^2 = 0.85$, $p = 0.078$); however, this relationship was almost entirely driven by the September measurements, which exhibited both comparatively low CWC values and very negative midday LWP values. If the September data is removed, we observe $R^2 = 0.12$, $p = 0.77$.

CWC was more consistently but still weakly correlated with predawn LWP for most dates across space but was significant in three of the four measurement dates compared to two of four for midday LWP (Fig. 5c). However, over time, predawn LWP was strongly correlated with CWC ($R^2 = 0.99$) with strong statistical significance ($p = 0.01$) (Fig. 5d), and not purely dependent on the September sampling date.

3.4. Using CWC to map water stress at the landscape level

Interestingly, topography was not strongly associated with any of the three leaf-level tree water stress metrics (LWC, midday LWP, or predawn LWP) (Fig. 6). Indeed, linear mixed effects models using all available topographic variables but omitting CWC exhibited little to no predictive capacity for LWC or LWP, with relatively low marginal R^2 values (values ranging from 0.029 to 0.056; see Table S2). The most parsimonious model predicting predawn LWP was a model that used only CWC and northness (Table S3), whereas the most parsimonious model predicting

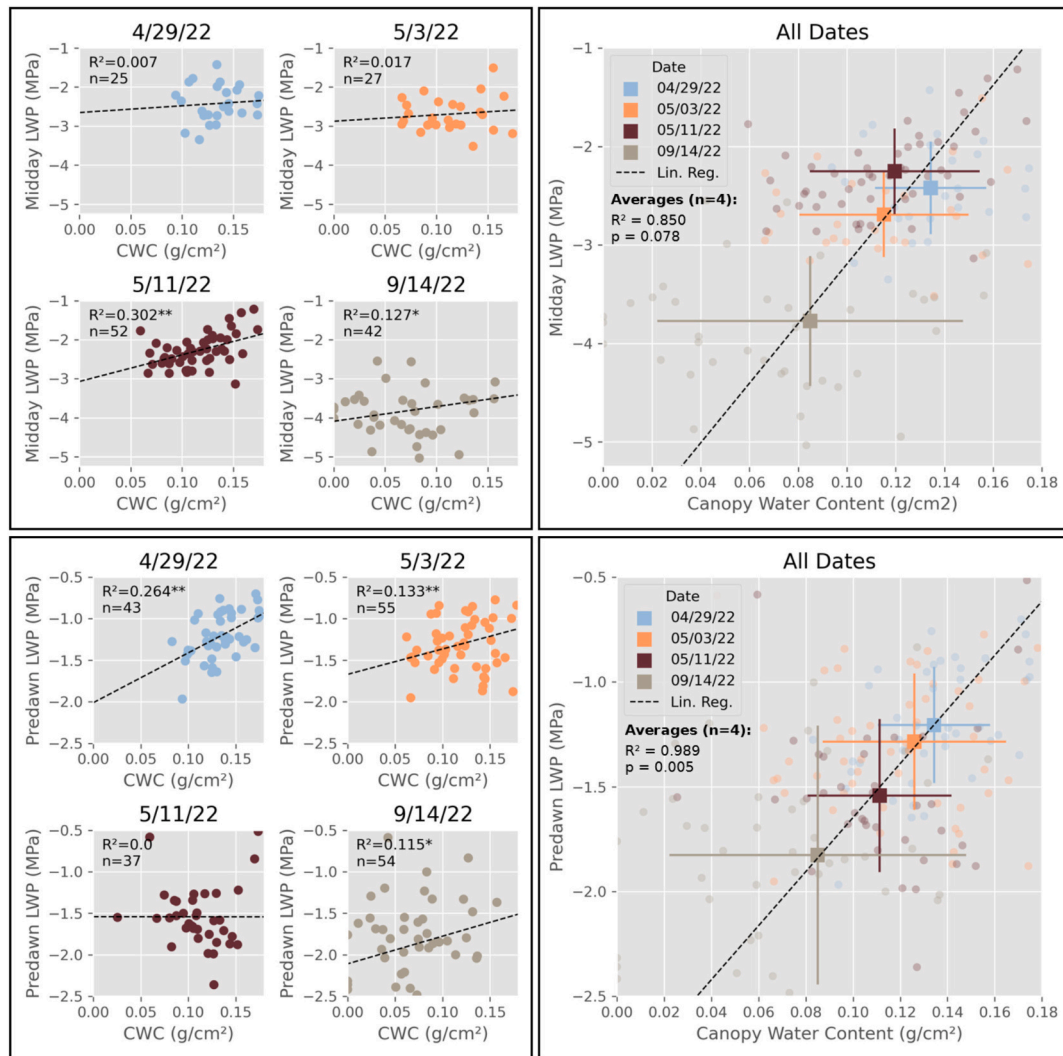


Fig. 5. Relationships between midday LWP and CWC across space vs. through time were similar to the relationships between LWC and CWC. (a) Scatterplots between midday LWP and CWC for four SHIFT dates with linear regressions. (b) Average midday LWP and average CWC plotted against each other for each of those four dates. (c-d) Same as a-b, but for predawn LWP. Here, * indicates results that were statistically significant at $p < 0.05$; ** indicates results that were statistically significant at $p < 0.01$.

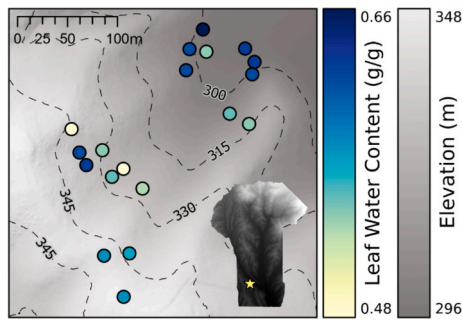


Fig. 6. Tree water status is not well predicted by elevation. Map of one study site used for this analysis which showcases variation in LWC for September across a 60 m elevational gradient. Absolute elevation is shown on contour lines and is measured in meters. As is visible in this example case, topography is not well correlated with tree water status, and even adjacent trees can exhibit very different LWC values.

midday LWP used all available topographic variables plus CWC. Models predicting midday LWP, with or without CWC, also had slightly higher marginal R^2 values than models predicting predawn LWP (Table S3). This indicates that, at least in our study system, topography exhibits less influence on soil water access than it does on atmospheric dryness or heat. This was unexpected, as Sedgwick Reserve contains multiple seasonal creeks and many small valley features that could provide topographic controls on subsurface water availability and study trees were explicitly selected to capture differences in topographically mediated soil water. However, the weak topographic relationships exhibited directionally consistent effects across all three response variables, with trees with a more hydrated water status (less negative predawn/midday LWP, higher LWC) at higher elevations and shallower, south-facing slopes. While the marginal R^2 values were low, conditional R^2 values were high across all response variables (where the most parsimonious models yielded values ranging from 0.456 to 0.781; see Table S2), indicating that the two included random effects (species and date) contributed substantially to overall model performance. Null models that used only random effects revealed that date was the dominant random effect for LWC and midday LWP (47.17% and 50.33% variance explained by date for LWC and midday LWP respectively with 26.25% and 6.26% variance explained by species), whereas species exhibited a moderate control on predawn LWP (27.7% variance explained by species with 18.52% variance explained by date).

Including CWC measurements consistently improved the linear mixed effects models, but overall model performance remained low. Adding CWC to the most parsimonious topographic models for all variables decreased AIC and increased marginal R^2 values when compared with models without CWC, indicating that CWC is overall additive when producing predictive models for tree water stress. In particular, adding CWC and interactions between CWC and topography to models predicting midday LWP resulted in a decrease in AIC of more than 13 (Table S2), which is strong evidence that CWC is usefully predictive of midday LWP, even if the overall predictive power is low. However, CWC was comparatively less useful in modeling LWC, a property to which it is more directly physically linked. For all three response variables, higher CWC was associated with more hydrated water leaves. Models that included interactions between topography and CWC yielded interaction terms that were mostly statistically insignificant with a few exceptions: For predawn LWP, the CWC effect weakened at higher elevations, and for midday LWP and LWC, this effect weakened on steeper slopes. There was also a statistically significant interaction between CWC and northness when predicting LWC, where the CWC effect weakened to the point of becoming negative on north-facing slopes.

4. Discussion

Here, we examined CWC as a diagnostic for tree water stress in an oak grass savanna. We saw that spatial variation in CWC across the landscape was much more strongly linked to LAI than LWC or midday LWP – that is, that CWC was more related to leaf quantity than leaf hydration. The one statistically significant relationship between LWC and CWC across space was a negative correlation – the opposite of what would be expected for a physically based relationship – suggesting that it was either random or driven by some underlying covariate not measured here. These results corroborate previous results in both modeling and observational frameworks (Baret et al., 1992; Bohn et al., 2020) that indicate that across space, CWC is best related to LAI and not LWC; that is, CWC is more sensitive to leaf area than it is leaf water. In contrast, over the time series, CWC was observed to decline coincident with a decline in LWC, and to a much lesser extent LAI (compare an average 28.8% decline in average LWC from April to September to an average 14% decline in LAI over the same time period). Indeed, LAI was much more variable across space than LWC, with a coefficient of variation 1.5–2 times that of LWC and LWC varied more strongly over time than LAI (Fig. S4). However, all temporal relationships were sensitive to the final data point collected in September; as this point represents a severe, end-of-season stress state following a damaging heat wave, care should be taken in interpreting how these results would translate to other ecosystems or other years.

At the landscape level, variations in tree water status were complex and did not strongly correspond with topography in our study system. Using a linear mixed effects model framework, we found that including CWC as a fixed effect and especially as interactions with topography substantially improved predictions of both LWC and LWP, though overall model performance was modest at best (all marginal R^2 s < 0.1). Interestingly, model performance was better for LWP than LWC, which has a more direct physical link to CWC. This suggests that the value of CWC in predicting midday LWP is not primarily derived from its sensitivity to differences and fluctuations in leaf water, but rather the underlying relationships between CWC and other tree water stress indicators, such as leaf thickness, dry matter content, leaf angle, or LAI, which are likely related to CWC through less mathematically straightforward relationships and may vary across differences in species compositions, genotypes, and/or legacies of disturbance or drought.

4.1. Underlying mechanisms behind CWC

CWC is an integrated measurement of leaf mass per area, the amount of water by mass in those leaves, and the total leaf area per ground area. However, the exact relationship between these three variables and measurements of CWC derived from imaging spectroscopy data is relatively unknown. In this study, we observed different CWC values depending on the wavelengths used for the Beer-Lambert inversion. We posit that the 1100–1300 nm wavelength range was most appropriate for our study because it showed stronger agreement with the field measurements; however, this may not hold in ecosystems with different canopy structures or LAI values. In this savanna ecosystem, much of the variability across wavelengths in the fit between the field measurements and the airborne CWC data was caused by low measurement sensitivity at wavelengths where the absorption coefficient of liquid water is small, but nonzero (especially the 850–1150 nm wavelength range). Previous research has also observed this type of measurement failure when calculating CWC using the Beer-Lambert method (Boving et al., 2025). In an ecosystem where low measurement sensitivity poses less of an issue (i.e., a high LAI ecosystem where CWC values are expected to be higher), it's likely that Fig. 2b could look very different.

Even presupposing that imaging spectroscopy provides accurate measurements of CWC, it is currently unclear how to best decompose CWC into its constituent components. Leaf-level spectroscopy studies have revealed a tight and linear relationship between LWC and Beer-

Lambert measurements of liquid water for individual, isolated leaves (Hunt and Rock, 1989). As we observe here, however, scaling relationships between leaf spectral measurements and above-canopy imaging spectroscopy data is not trivial (Asner and Martin, 2008; Gara et al., 2019). The relationship between LWC, LAI, and CWC as measured by an imaging spectrometer is likely mediated by other leaf traits and canopy structural properties that impact the dynamics of light scattering within the canopy, such as leaf mass per area or leaf angle distribution. Here, we have attempted to control for these auxiliary factors by focusing on two *Quercus* species that have overall similar canopy structures and understory compositions while leveraging a period of phenological stability. This should isolate the majority of the variation in our airborne measurements of CWC to the factors that vary the most systematically across our study dataset: LAI and LWC.

The extremely low CWC measurements observed in the final September data collection flight are likely due to declines in LAI associated with a fall heat wave (Fig. 1C). Elevated temperatures late in the dry season led to early leaf senescence for upper canopy leaves in *Q. douglasii* (the most abundant species in the domain) which drove a decline in CWC. If this September measurement is excluded, the relationship between average CWC and average LWC remains relatively strong, but exhibits reduced statistical significance ($R^2 = 0.811$, p -value = 0.286). Therefore, the research presented here provides more evidence that CWC is useful for differentiating between these extreme early-season and late-season states, and only measured evidence that CWC is sufficiently sensitive to monitor smaller, short-term changes in water stress that occur on the timescale of days or weeks. A longer or denser time series could provide more insight into the stability and significance of this relationship.

These dynamic changes in both LAI and LWC highlight an important caveat when using CWC for monitoring tree water stress: Underlying correlations between tree stress responses at the leaf and canopy structural level can mediate the CWC/LWC relationship which – depending on the intended use case – could either be helpful or detrimental. Underlying correlates like this could serve to amplify the CWC signal associated with water stress, providing stronger detection sensitivity and aiding in the development of empirical methods for mapping water stress. Alternatively, they could serve as a confounding variable when a specific link between CWC and some certain ecological measurement is desired. In either case, care must be taken when scaling the use of a CWC data product across a larger region, particularly when operating out of sample from existing measurements.

4.2. Implications for mapping tree water stress at the landscape level

Relating CWC measurements to ecologically relevant metrics of tree water stress, such as risk of tree mortality, is an innately challenging problem. Here we propose using LWP as a functional liaison between CWC data and actual water stress. Midday LWP has a strong physical relationship with LWC and is influenced by predawn LWP and atmospheric dryness; therefore, it is a reasonable hypothesis that midday LWP and/or predawn LWP will vary in a fairly well-constrained way with CWC. Our results offer moderate support for this hypothesis: Over time, LWC was more strongly correlated with CWC than midday LWP, but the difference in explanatory power was relatively small (compare LWC: $R^2 = 0.89$, $p = 0.058$ to LWP: $R^2 = 0.85$, $p = 0.078$). Furthermore, mixed effects models yielded a higher marginal R^2 value when predicting midday LWP than when predicting LWC. This suggests that it may be feasible to empirically relate changes in CWC directly to changes in LWP, rather than relating CWC to LWC and using that to draw inferences about LWP (Boving et al., 2025). This is particularly reasonable in ecosystems like Sedgwick Reserve, where a relatively narrow range of tree species is being considered and the LWC-LWP relationship is relatively consistent.

Interestingly, CWC showed the best (though still modest) predictive power for predawn LWP across space on most dates. In fact, CWC was

more predictive of predawn LWP than any of the considered topographic variables, potentially indicating that the trees are accessing deep water sources not correlated with topography, such as water stored in rock fractures (McCormick et al., 2021) or near the water table. We hypothesize that the relationship between CWC and predawn LWP is due to an underlying correlation between predawn LWP and LAI (Fig. 7) – that trees with more soil water access grow more numerous and larger leaves, resulting in higher LAI and therefore higher CWC values. This may also explain the observed interaction between CWC and elevation: The depth of the dry soil layer separating trees from deep water sources likely increases with elevation, which would result in a weaker relationship between predawn LWP and LAI – and therefore, CWC – at high elevations. This provides another example of how CWC may be useful as a tool for indirectly monitoring water status in trees, even in cases where only one imaging spectroscopy dataset is available (e.g., most airborne campaigns). It also supports prior results from giant sequoia (*Sequoiadendron giganteum*) suggesting that CWC variation is typically tied to drought responses through leaf area changes rather than water potential-driven changes in leaf hydration per se (Martin et al., 2018). If this is the case, these responses could be expected to vary substantially across different species and environmental conditions due to differences in water use strategy, rooting depth, legacy responses to drought, and other factors that vary at the landscape scale.

Overall, the stronger predictive power of CWC for midday and predawn LWP (as opposed to LWC) could point to CWC providing a proxy for the overall water stress state of a pixel by synthesizing information on canopy structure, leaf hydration, and other optical effects that could be correlated with LWP. The high bias observed in the CWC measurement relative to the field measurements indicates some level of mismatch between the actual water per unit area stored in leaves and a remote CWC measurement, likely caused by understudied multiple scattering interactions between different materials in the canopy. This mismatch could contribute to or even drive the relationships between CWC and LWP observed here. In this counterintuitive way, CWC could be valuable as a proxy for LWP despite weak relationships with physical canopy traits: CWC as a composite measurement of many canopy properties may be more predictive of tree water stress than a direct measurement of leaf water per area would be on its own.

Finally, these measurements occurred during a series of much drier than average years for Santa Barbara County. This comes with both challenges and opportunities. The dry conditions resulted in a higher level of baseline stress (e.g. lower starting LWP, even at leaf out) at the

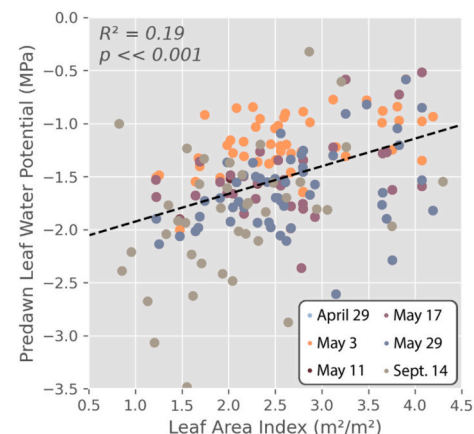


Fig. 7. LAI and predawn LWP are related to each other, which could explain the strong utility of CWC for predicting predawn LWP: Trees with more subsurface water availability grow more or larger leaves, which results in an apparent sensitivity of CWC to predawn LWP. Dates indicate the predawn LWP measurement date, and LAI data is sourced from the closest single LAI measurement in time for that tree. The dotted line and reported R^2 and p value show a linear regression across all trees and all dates.

beginning of the SHIFT measurements and less of a potential gradient in LWP/LWC over the course of the spring drydown before trees reached physiological thresholds that induced full stomatal closure compared to a normal or wetter than average year with higher water availability. While both oak species in this dataset are understood to be fairly tolerant of low LWPs, many of the observed LWP values are extreme even relative to the trees' own tolerances. The turgor loss point (sometimes called the wilting point), or the LWP value below which leaf function declines, is estimated to be approximately -2.68 MPa for *Q. agrifolia* and approximately -3.37 MPa for *Q. douglasii* (Fig. S5) and the P50, or LWP that causes 50% leaf xylem embolism, is around -4.47 MPa for *Q. agrifolia* and -4.45 MPa for *Q. douglasii* (Skelton et al., 2018). Thus, our data indicate that many trees in this study period were operating at or beyond their hydraulic tolerances during this dry year. As a result, there was potentially less LWP variation throughout the spring to summer window because LWP started at a lower value than average. This may have reduced the observable gradient in hydration that could be related to CWC. However, dry conditions likely also maximized the possible observable spatial variability in LWP/LWC – heavy rains would have refilled soil water supplies for all trees in the study dataset, resulting in more uniform water stress conditions. Ideally repeat measurements over years with varying water availability will help to better constrain how CWC and LWC jointly vary within an ecosystem over years with strong interannual variability in water availability, and across different ecosystem types.

4.3. Looking forward to the next generation of imaging spectrometers

Our results suggest that having timeseries of hyperspectral data are critical for leveraging CWC data for water stress monitoring. CWC was not predictive of LWC or midday LWP across space in general, most likely due to underlying confounding factors, such as variation in LAI, leaf mass per area, and other plant structural attributes. However, declines in CWC over time were associated with declines in LWP and LWC for a time period where LAI was relatively stable. A single CWC measurement in time poses an under constrained problem, with only one measurement that varies with three unknowns (leaf quantity, leaf mass per area, and leaf hydration), each of which relate to a wide array of leaf and canopy traits. Many of these leaf and plant traits have been linked to imaging spectroscopy data at the leaf and/or canopy level; however, the best way to produce maps of these relevant traits (especially leaf mass per area) remains an area of active research. Adding more measurements in time whereby the same tree can be reimaged, controlling for most structural variation, offers an opportunity to resolve this problem without direct measurement of those other traits, and serves as an example of how imaging spectroscopy data collected in time series can capture information not observable from single scenes.

This time series approach to CWC monitoring may be more feasible in coming years. For most of the history of imaging spectroscopy, data have been collected by airborne instruments, making time series monitoring challenging to collect, particularly in more remote study areas. However, with the advent of spaceborne imaging spectrometers, such as the Earth Surface Mineral Dust Source Investigation (EMIT), the Environmental Mapping and Analysis Program (EnMAP), Carbon-Mapper, and others, time series monitoring of CWC is likely to become increasingly feasible for more locations around the world. In particular, CWC is one of the planned products for the NASA Surface Biology and Geology (SBG) designated observable that was called for in the 2017 decadal survey (Cawse-Nicholson et al., 2021). Should SBG be launched, CWC data will be available on a global basis on a roughly two-week revisit time, barring blockage by clouds.

Further work is needed to determine how and when CWC can be used for monitoring water stress in trees. Our results suggest that CWC changes with changing leaf water in this particular ecosystem and potentially other savanna ecosystems; however, both this study site and the SHIFT dataset were in many ways exceptional and the results may

not transfer to other sites. For example, it is reasonable to speculate that intra- or inter-species differences in vegetation structure, such as changes in leaf habit, woody biomass, leaf mass per area, or leaf angle, would change the sensitivity of CWC to fluctuations in leaf water. This study benefitted from previous field research into how some of these potential confounding factors (especially leaf mass per area and phenology) could influence our CWC measurements (Boving et al., 2025). However, this valuable pre-requisite information that informed our study design is not available everywhere and could pose a major barrier to entry in transferring this approach. Furthermore, the high spatial resolution of the SHIFT dataset (5 m) enabled us to select only pixels that had a high tree fractional cover; at coarser spatial resolutions, including data collected by current and upcoming spaceborne spectrometers (which may be 30–60 m), mixed pixels are likely to be unavoidable. This would add another dimension of complexity to the problem and likely some reduced measurement sensitivity for variations in LAI or LWC in the trees specifically. Coarser spatial resolution data would likely also sample different individuals or even different species, further complicating the retrieval of any specific metric. Future research comparing CWC to field measurements across a wider array of vegetation types and structures may illuminate methods of overcoming this issue.

Another promising direction for future work concerns the applicability of the Beer-Lambert model. CWC as retrieved from the AVIRIS-NG imagery exhibited a strong positive bias – nearly an order of magnitude – when compared to the field-based estimates of leaf water per unit area. We hypothesize that this is due to a mismatch between the average amount of water light passes through in a tree canopy and the actual leaf water per unit area, and that some water is effectively “double measured” according to the assumptions of the Beer-Lambert retrieval. This mismatch could be occurring at the leaf level (i.e., within-leaf scattering increasing the path length of light (Allen et al., 2026)), at the canopy level (i.e., within-canopy scattering causing a given ray to travel through the same leaf twice, or a larger proportion of light that passed through multiple leaves being returned to the sensor), or both. Both effects would stem from optical interactions with structural matter in the leaf or canopy that is not accounted for in the Beer-Lambert model, such as leaf dry matter and branches. At present, these effects are poorly constrained, and it remains unclear how much they contribute to variability in CWC across time or space, or how they might be measured in the field. If these effects are both significant and strongly variable, they could contribute to error in the relationship between CWC and LAI/LWC. More widely available leaf and imaging spectroscopy data across more diverse vegetation types will be important if we are to advance our understanding of how these optical interactions contribute to a CWC measurement and, therefore, how best to calculate and interpret CWC data across ecosystems.

Finally, this dry environment provided confidence that all liquid water measured by the spectrometer was contained in tree leaves, whereas in more humid environments it has been demonstrated that CWC is strongly affected by liquid water on the leaf surface (Xu et al., 2021). As more high-frequency CWC datasets become available, studies of how CWC varies with plant water stress will become feasible for a wider variety of locations, which will likely continue to reveal relevant confounding factors that apply in other ecosystems.

4.4. Conclusions

Our results suggest that the relationship between CWC and both LAI and LWC is more complex than previously thought. We found that, while the relationship between CWC and LAI was stronger and more consistent than the relationship between CWC and LWC, it was still relatively weak and somewhat variable. This suggests that there is substantial variability in CWC that is not explained by either LWC or LAI – that is, CWC is not simply an integrated measurement of leaf water per unit area. Furthermore, the relationship between CWC and predawn LWP was

substantially stronger than the relationship between CWC and LWC or midday LWP. While this bodes well for the use of CWC as a tree water stress monitoring tool, it also underscores gaps in our knowledge in how CWC (and canopy spectral measurements as a whole) responds to changing environmental conditions. Access to data in time series can, in part, provide tools for overcoming this barrier, by providing multiple measurements in time across which to construct an empirical relationship – but without a stronger mechanistic relationship, we will struggle to develop models that effectively scale across large regions or between ecosystems.

As droughts become increasingly frequent and severe, identifying which trees and ecosystems are most vulnerable to drought-induced mortality will become an increasingly important priority across all land management scales. Our work here suggests that CWC exhibits strong potential as a tool for monitoring drought status in open woodlands or savanna-like ecosystems; however, more research is needed to understand how these results scale across different ecosystems with variable species compositions and structural traits. At present, the SHIFT dataset is unique; however, in a future where imaging spectroscopy time series are available on regional or global scales, this study and future work that builds on it will contribute to development of transferable and scalable methods of mapping tree water stress and help push the field of remote sensing towards large-scale tree vulnerability monitoring.

CRedit authorship contribution statement

Jean Allen: Writing – original draft, Visualization, Methodology, Funding acquisition, Formal analysis, Conceptualization. **Indra Boving:** Writing – review & editing, Methodology, Investigation, Data curation. **Leander D.L. Anderegg:** Writing – review & editing, Resources, Methodology, Investigation, Funding acquisition, Conceptualization. **Philip G. Brodrick:** Writing – review & editing, Conceptualization. **K. Dana Chadwick:** Writing – review & editing, Resources. **Dar Roberts:** Writing – review & editing, Methodology. **Anna T. Trugman:** Writing – review & editing, Resources, Methodology, Funding acquisition, Conceptualization.

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Declaration of competing interest

Jean Allen reports financial support was provided by NASA. Anna T. Trugman reports financial support was provided by Gordon and Betty Moore Foundation. Anna T. Trugman reports financial support was provided by National Science Foundation. Leander D. L. Anderegg reports financial support was provided by National Science Foundation. Anna T. Trugman reports financial support was provided by California Department of Forestry and Fire Protection. Leander D. L. Anderegg reports financial support was provided by California Department of Forestry and Fire Protection. Anna T. Trugman reports financial support was provided by US Department of the Interior. Leander D. L. Anderegg reports financial support was provided by US Department of the Interior. Given his role on Editorial Board, DR had no involvement in the peer review of this article and had no access to information regarding its peer review. Full responsibility for the editorial process for this article was delegated to another journal editor. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.rse.2026.115374>.

Data availability

SHIFT L2A Surface Reflectance data (Brodrick et al., 2024) can be found on the Oak Ridge National Laboratory Distributed Active Archive Center (ORNL DAAC). Subsets of the SHIFT surface reflectance data used in this research, field data, and code used for data analysis can be found on GitHub at https://github.com/jean-allen/SHIFT_CWC.

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